where α is the restriction of ν to scalars, has the character

$$\chi\begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} = (q-1)\alpha(x),$$

$$\chi\begin{pmatrix} x & 1 \\ 0 & x \end{pmatrix} = -\alpha(x),$$

$$\chi\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} = 0,$$

$$\chi\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix} = -\nu\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix} - \nu^q\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix}.$$

In all that follows, we will have $\nu^q \neq \nu$.

The following two lemmas will establish that the inner product of this character with itself is equal to 1 and that its value at 1 is positive. As we know from Lemma 5.7.2, these two properties imply that it is the character of an irreducible representation of G.

Lemma 5.25.3. *Let χ be the character of the virtual representation defined above. Then*

$$\langle \chi, \chi \rangle = 1$$

and

$$\chi(1) > 0.$$

Proof.

$$\chi(1) = q(q+1) - (q+1) - q(q-1) = q - 1 > 0.$$

We now compute the inner product $\langle \chi, \chi \rangle$. Since $\alpha(x)$ is a root of unity, this will be equal to

$$\frac{1}{(q-1)^2 q(q+1)}\big[(q-1)\cdot(q-1)^2\cdot 1 + (q-1)\cdot 1\cdot(q^2-1) + \frac{q(q-1)}{2}\cdot\sum_{\zeta \text{ elliptic}}(\nu(\zeta)+\nu^q(\zeta))\overline{(\nu(\zeta)+\nu^q(\zeta))}\big].$$

Because ν is also a root of unity, the last term of the expression evaluates to

$$\sum_{\zeta \text{ elliptic}}(2+\nu^{q-1}(\zeta)+\nu^{1-q}(\zeta)).$$

Let's evaluate the last summand.